Jonnadula Mohit Sai

AI/ML Engineer • Gen AI Engineer • Full-Stack Developer • AWS Certified Cloud Practitioner
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PROFESSIONAL SUMMARY

Final-year Computer Science student (2026) specialising in AI/ML and backend systems. Built production-grade GenAI applications including a multi-modal RAG pipeline, edge AI voice assistant, and a custom Transformer TTS engine trained from scratch. Experienced deploying scalable systems on AWS across 3 internships. Seeking Software Engineer / AI Engineer / Gen AI roles.

EXPERIENCE

Full-Stack Engineer Intern (Remote, Part-time) Jan 2025 – Oct 2025
SSIT Solutions

- Delivered a production Secure Exam Portal (MCQ + coding challenges, RBAC) for an enterprise client — transitioned from 3-month training to paid engineering after demonstrating capability.
- Engineered REST APIs in Node.js/Express achieving 20% reduction in server response time; deployed auto-scaling microservices on AWS EC2 + S3 serving production traffic.
- Optimised React.js dashboards for data-heavy views; implemented JWT auth and role-based access control securing multi-role student and admin workflows.

Python Developer Intern May 2024 – Jul 2024
Edubot Technologies (Hybrid)

- Automated dataset preprocessing pipelines, cutting manual data-cleaning effort by 40% and accelerating model training cycles.
- Integrated ML models into a live educational platform, engineering real-time data-processing scripts handling concurrent user streams.

Web Development Intern Jan 2024 – Mar 2024
Biztron Softech (On-site)

- Built responsive cross-browser UIs and resolved critical frontend performance bottlenecks, improving page load consistency across devices.

PROJECTS

Multi-Modal RAG Assistant — "Dual-Brain" [Demo] | *Python · Mistral-7B · Qwen2-VL · FAISS · LangChain*

- Designed a retrieval system that intelligently routes text queries to Mistral-7B and diagram-based queries to Qwen2-VL — achieving ~30% higher accuracy on visual questions vs a text-only baseline.
- Implemented FAISS vector indexing with context-aware chunking, reducing query latency by ~35% and significantly reducing hallucinations in multi-modal responses.

Edge AI Voice Assistant — "The Guardian" [Demo] | *PyTorch · ONNX Runtime · Whisper · SmolLM2 · SpeechT5*

- Built a fully offline voice agent (Whisper STT → SmolLM2 → SpeechT5 TTS) eliminating cloud dependency — optimised via ONNX Runtime, achieving ~60% lower inference latency vs cloud-based alternatives.
- Quantised model weights and optimised the ONNX inference graph, enabling real-time voice response on standard consumer hardware with no GPU required.

Custom Transformer TTS Engine — "MiniTTS" [Demo] | *PyTorch · Seq2Seq Transformers · Streamlit*

- Implemented a full encoder-attention-decoder Transformer TTS architecture from scratch in PyTorch — trained on 13,000 audio clips without any pre-built TTS library, deployed via Streamlit for real-time inference.
- Overcame black-box limitations of existing TTS models by building a fully interpretable architecture with controllable prosody.

Postit — Social Media Platform [GitHub] [Demo] | *MongoDB · Express · React · Node.js · AWS · JWT*

- Engineered a full-stack social platform with real-time feeds, JWT auth, follow system, and sub-100ms API latency — deployed on AWS handling concurrent users.

More live demos: Industrial Technician AI | Foley Artist AI

TECHNICAL SKILLS

AI / Gen AI: PyTorch, ONNX Runtime, Hugging Face, LangChain, FAISS, Whisper, Seq2Seq Transformers, Computer Vision, NLP, Prompt Engineering, Streamlit, Gradio
Backend & Cloud: Python, Node.js, Express.js, FastAPI, REST APIs, System Design, AWS (EC2, S3, Lambda), Docker, CI/CD, Git, Java.
Frontend & DB: React.js, Next.js, JavaScript ES6+, TypeScript, Tailwind CSS, Redux, MongoDB, PostgreSQL, SQL, Firebase, HTML

EDUCATION

B.Tech — Computer Science & Engineering CGPA: 8.07 / 10
Lakireddy Baliredy College of Engineering, Vijayawada 2022 – 2026
Certifications: AWS Certified Cloud Practitioner (2024) • IBM Machine Learning • HarvardX Data Science

ACHIEVEMENTS

- Hackathon Winner — 24-hr AI Engineering Challenge, ranked #1 among all competing teams.
- Deployed 5+ production AI systems with live demos on Hugging Face Spaces, representing 2,000+ lines of original code.
- AWS Certified Cloud Practitioner -earned during 3rd year Btech